

Network Security
(SS/ES ZG513)

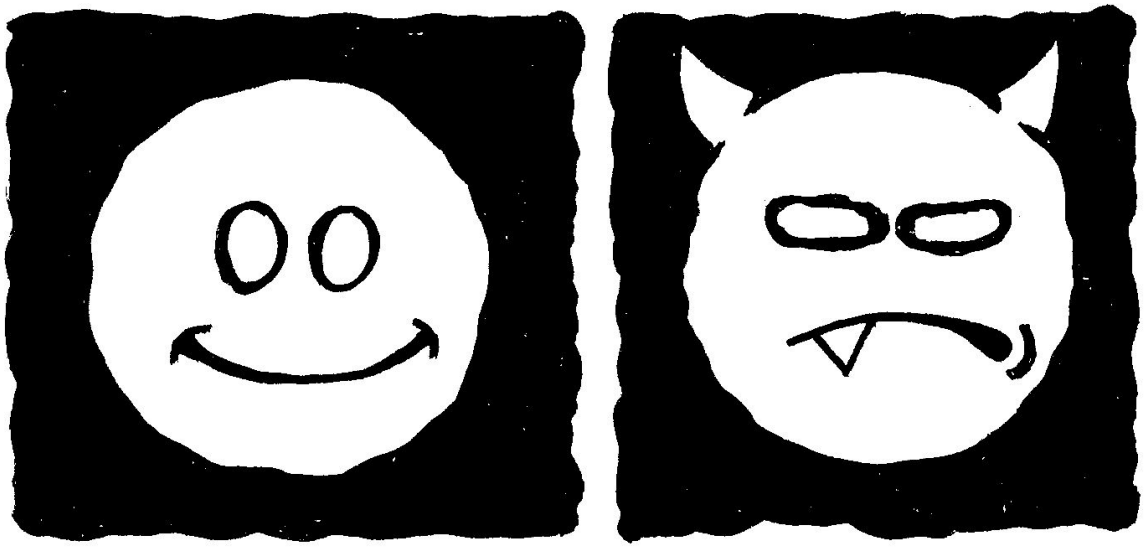
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Contact Session – I

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Why Network Security



Protect Good from Bad

- "The art of war teaches us to rely not on the likelihood of the enemy's not coming, but on our own readiness to receive him; not on the chance of his not attacking, but rather on the fact that we have made our position unassailable"

Definitions

- Data Security: measures that are applied to protect digital privacy and to prevent unauthorized access to computers, databases, websites, etc.
- Computer Security: generic name for the collection of tools designed to protect data and to thwart the hackers.
- Network Security: measures to protect the data during their transmission.
- Internet Security: measures to protect data during their transmission over a collection of inter-connected networks.

Background

- Information Security requirements keeps on changing.
- Traditionally provided by physical and administrative mechanisms.
- Computer use requires automated tools to protect files and other stored information.
- Use of networks and communications links required measures to protect data during transmission.

Network Security: Some Facts

- Federal government has suffered 680% increase in network/cyber security breaches in the past six years.
- Governments, not hackers, are most likely to launch cyber attacks.
- More than 600,000 accounts are compromised every day on Facebook alone.
- National Nuclear Security Administration records 10 million attempted hacks a day.
- US Navy receives 110,000 attacks per hour.
- Every second 18 adults suffer cybercrime (1.5 million/day).
- Global spam rate is 68%. Of these 61% are adult/dating messages, 28% are pharmaceutical.

Course Objectives

- Comprehensive course on Network Security: Includes both theory and practice.

- Information security is an important area of information technology and this course on Network Security helps the students to understand the important security aspects in the computer networks, cryptographic techniques and protocols to implement the security goals.

- The course provides a necessary review of mathematical concepts to implement different cryptographic techniques then provides a deeper dive to the field of cryptography and methods to implement them.

Course Objectives

- To understand the important security goals in the networks - Confidentiality, Integrity, Authentication, Non-repudiation and Availability using cryptographic techniques.

- The course provides a top-down approach to explore the security implementations in important three layers of Internet protocol stack i.e. Application, Transport and Network layer.

- The course consolidates and sums up the learning taking few case studies and examples from latest trends and industry deployments.

What Will You Learn: A Sample

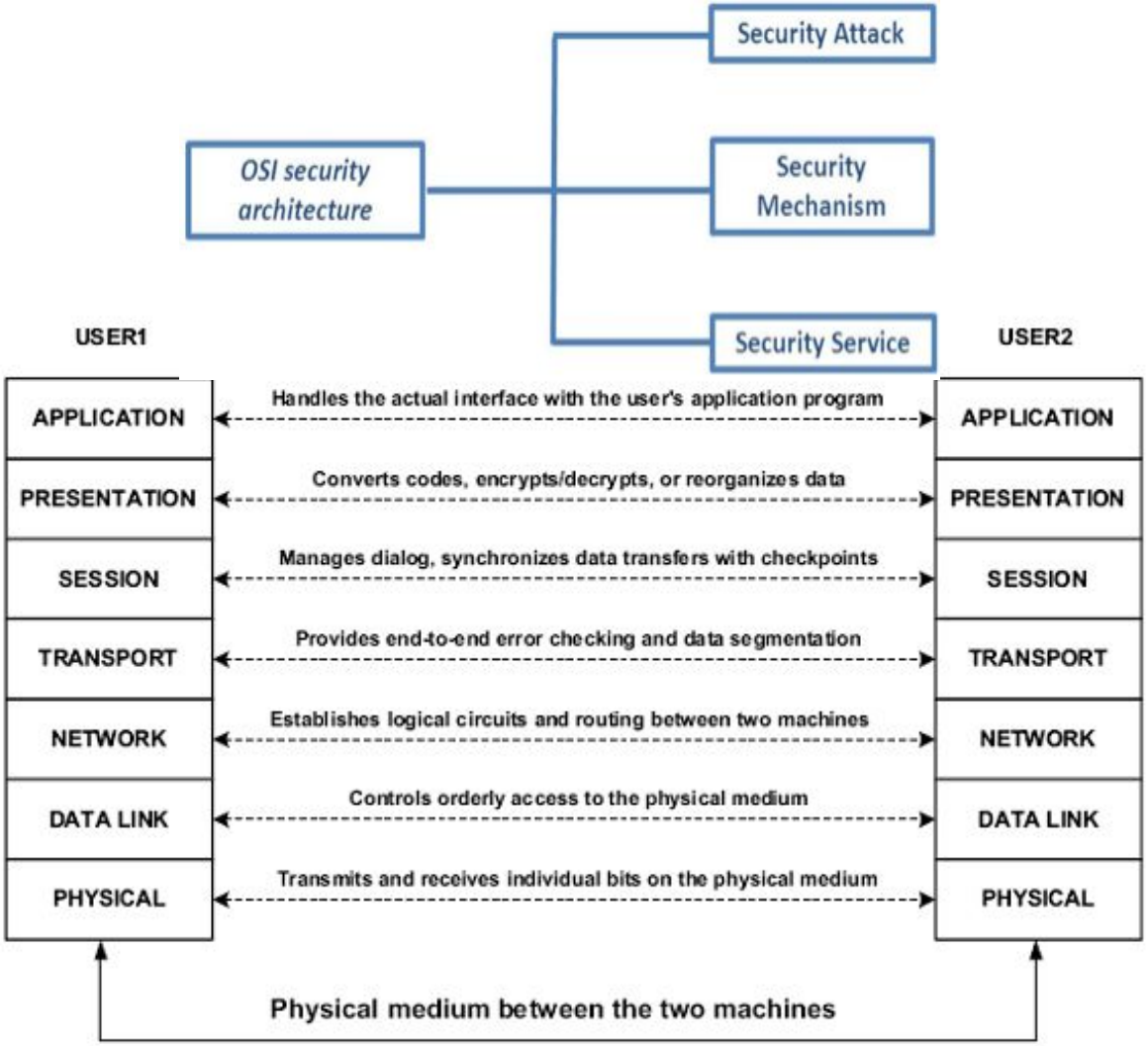
- Cryptography:
  - Encryption techniques
  - Hashing techniques and MAC
  - Digital Signature and Certificates
  - Some Protocols: DHKE, Kerberos etc.
- How to exchange security keys over public network?
- How you can sign a message confirming that you sent it? You can't deny it in a court of law.
- What are certificates? How you confirm that you are talking to "XYZ"?
- Application Layer Security: Email
- Transport Layer Security: SSL/TLS, SET
- Network Layer Security: IPSec

What Will You Learn: A Sample

- Text Book:
  - Stallings William: Cryptography and Network Security - Principles and Practice, Pearson India, 7th Edition, 2017 or later.
- Reference Books:
  - Forouzan B A, Mukhopadhyay Debdeep : Cryptography and Network Security, McGraw, Hill, 3rd Edition, 2015 or later.
  - Bruce Schneier, Applied Cryptography: Protocols, Algorithms And Source Code In C 2nd Edition (Protocols, Algorithms and Source Code in C), Wiley India, 2nd Edition
  - Kurose James F and Keith W. Ross: Computer Networking: A Top-Down Approach, Pearson India, 5th Edition, 2012 or later.

OSI Security Architecture

- The Open Systems Interconnection (OSI) security architecture provide a systematic framework for defining security attacks, mechanisms and services.
- Network engineer has no choice but to be security conscious and vice-versa.
- No matter how well we secure our hosts, we are still vulnerable.

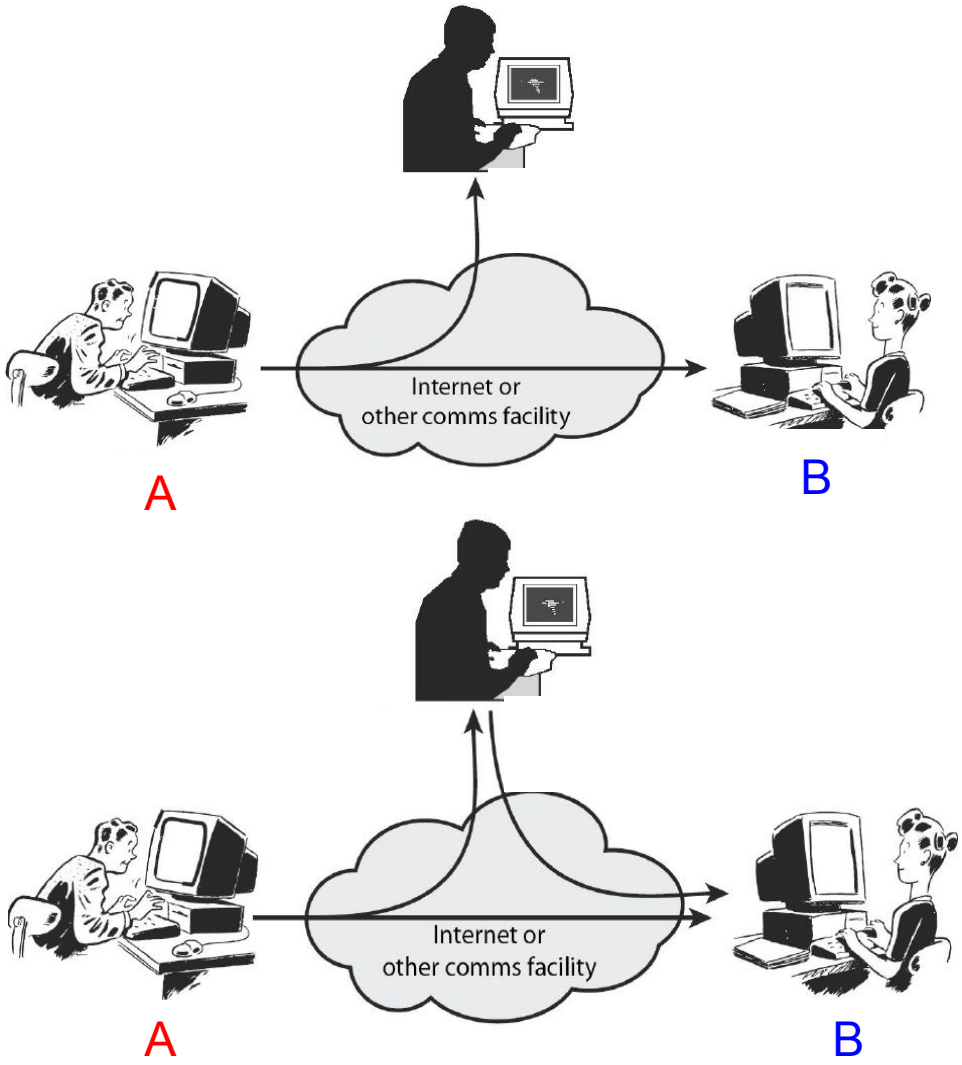


Aspects of Security

- Security Level
- Security Attacks
- Security Truism
- Security Policies
- Security Services
- Security Mechanism
- Security Response Teams

Security Attacks

- Any action that compromises the security of information owned by an organization
- Have a wide range of attacks
- Generic types of attacks:
  - Passive (Interception)
  - Active (Interruption, Modification, Fabrication)



Security Truism

- An integral part of the original design, no more privileges than those necessary to do the job, and make security arrangement simple as simple, but no simpler.
- Keep the level of all your defences at about the same height and in layers as attacker doesn't go through security, but around it (A chain is only as strong as its weakest link).
- No absolute security, bad idea to rely on Security through Obscurity.
- Security is always a question of economics; don't underestimate the value of your assets.
- If you do not run a program, it does not matter if it has security holes (program/protocol is insecure until proven secure).

Security Policies

- Which is not expressly prohibited is permitted and *vice-versa.*
- What resource are you trying to protect.
- Who is interested in attacking you.
- How many security can you afford.
- What action will be taken when security is compromised in the organization.
- What action will be tolerated and what will not.

Security Services

- Security Services implement security policies and are implemented by security mechanisms.
- Enhance security of data processing systems and information transfers of an organization.
- Intended to counter security attacks.
- Using one or more security mechanisms.
- Often replicates functions normally associated with physical documents, e.g.
    - Have signatures, dates, need protection from disclosure, tampering or destruction, be notarized or witnessed, be recorded or licensed.

Security Services (X.800)

- **Data Confidentiality:**
  - Protection of data from unauthorized disclosure.
- **Authentication:**
  - Assurance that the communicating entity is the one claimed.
- **Integrity:**
  - Assurance that data received is as sent by an authorized entity.
- **Non-repudiation:**
  - Protection against denial by one the parties in a communication.
- **Access Control:**
  - Prevention of the unauthorized use of a resource.

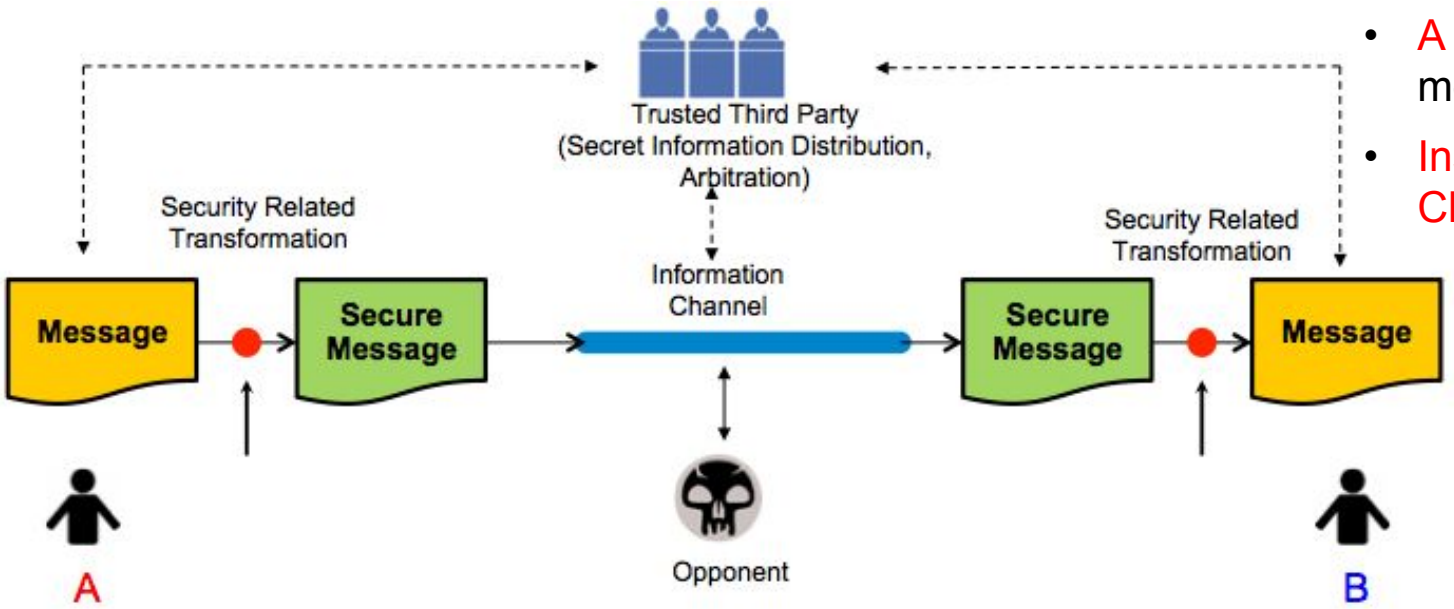
Security Mechanism

- Feature designed to detect, prevent, or recover from a security attack.
- No single mechanism that will support all services required.
- However, one particular element underlies many of the security mechanisms: Cryptographic Techniques.
- Specific Security Mechanism (X.800): Encipherment, Hash, MAC, Digital Signatures, Traffic Padding, Routing Control, Notarization.
- Pervasive Security Mechanism (X.800): Trusted Functionality, Security Label, Event Detection, Security Audit Trails, Security Recovery.

Security Response Teams

- Computer Emergency Response Team/ Coordinate Centre (CERT/CC)
- Defense Data Network Security Coordination Centre.
- NIST Computer Security Response and Response Clearinghouse.
- Department of Energy Computer Incident Advisory Capability.
- NASA Ames Computer Network Security Response Team.

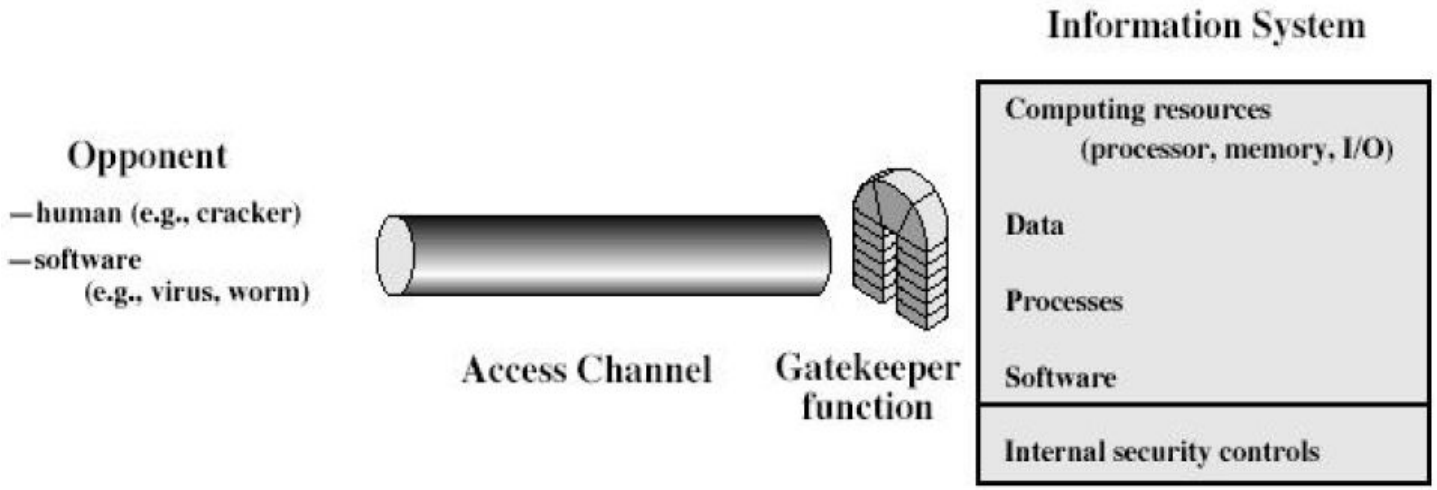
Model for Network Security



- A likes to send a message to B securely
- Insecure/ Secure Channel

- Using this model requires us to:
  - Design a suitable algorithm for the security transformation.
  - Generate the secret information (keys) used by the algorithm.
  - Develop methods to distribute and share the secret information.
  - Specify a protocol enabling the principles to use the transformation and secret information for a security service

Model for Network Security



- Kinds of Threats:
  - Information access threats (intercept/modify data on behalf of users who should not have access to the data)
  - Service threats (exploit service flaws in computers to inhibit use by legitimate users).
- Using this model requires us to:
  - Select appropriate gatekeeper functions to identify users
  - Implement security controls to ensure only authorised users access designated information or resources.
- Trusted computer systems may be useful to help implement this model.

Zero and Mutual Trust

- Mutual Trust is primarily two-way authentication for network communications (mTLS) and is a basis of Zero trust security architecture and operate on the principle of "Never trust, always verify".

- Principle of zero trust is perfectly aligned with mTLS, as it ensures all users, devices, and applications are authenticated regardless of their location (inside or outside the network perimeter).

- Mutual trust shall have strong defense against MITM attacks, spoofing, impersonation, and credential-stuffing attacks so that an attacker cannot easily forge either side communication/certificates.

- Requires managing a PKI to issue, distribute, and revoke certificates. It adds operational complexity compared to simple password-based methods, the use of automation tools, private CAs.

Cryptography: Terminology

- Cryptography: science of secret writing with the goal of hiding the meaning of the message.
- Cryptanalysis: art and science to break the cryptosystem.
- Encryption: method of transforming data (x) into an unreadable format.
- Plaint-text: message/data before encryption.
- Cipher-text: message/data after encryption.
- Decryption: method to get back the ‘x’ from ‘y’.

Cryptography: Terminology

- Cipher/EA: set of rules/procedures that dictates how to encrypt/decrypt the data.
- Key: values used in encryption/decryption.
- Key space: range of possible values used to construct keys.
- Key clustering: when two different keys generate the same ‘y’ from the same ‘x’.
- Work factor: estimated time and resources to break a cryptosystem. No system is unbreakable.

Steganography

- Hides secret message in other message.
- Security through obscurity.
- Does not attract attention, while cryptography draw attention.
- Provide secrecy, while cryptography provides privacy.
- Can be used where crptography is not allowed.
- Supplements cryptography.

Keys Ideas of Encryption

- Kerckhoffs's Principle.
- Confusion
- Diffusion

Breaking an Algorithm

- Total Break
- Global Deduction
- Instance (Local) Deduction
- Information Deduction

Cryptanalysis Attacks

- Ciphertext only attack
- Known plaintext attack
- Chosen plaintext attack
- Adaptive chosen plaintext attack
- Chosen ciphertext attack
- Chosen key attack
- Rubber hose cryptanalysis

Thanks!!!

Queries?